

Variational inference

Consider a simple Bayesian Gaussian mixture model:

$$\pi \sim \text{Dir}(\alpha)$$

$$\mu_j \sim \mathcal{N}(m_0, I \lambda_0)$$

$$c_i \sim \text{Cat}(\pi)$$

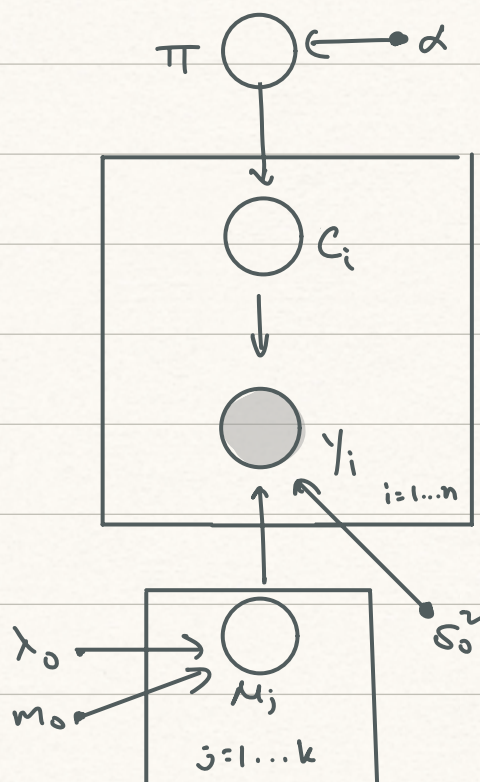
$$y_i \sim \mathcal{N}(\mu_{c_i}, \sigma_0^2)$$

$$\theta \equiv \{\mu_{1:k}, \pi\}$$

$$\eta \equiv \{\alpha, m_0, \lambda_0, \sigma_0^2\}$$

In lecture 6 we used the EM algorithm to do MAP inference in this model. Today we will be interested in **full Bayesian posterior inference over all variables**.

Here's the graphical model:



$$p(c, \theta \mid y, \eta)$$

The full posterior is intractable (as before). One thing we could do is run a Gibbs sampler to approximate the posterior with samples. This would be easy to implement since all the complete conditionals are closed-form. However MCMC has some downsides, such as being a stochastic algorithm, and potentially taking a long time to collect many samples.

Variational inference is an alternative. **It turns posterior inference into optimization.**

We begin by setting up a **variational distribution over all variables (i.e., over cluster assignments and parameters)**. We will want this distribution to approximate the posterior.

$$q(c, \theta) \approx p(c, \theta \mid y, \eta)$$

The variational distribution will have its own **variational parameters**

$$q(c, \theta) \equiv q(c, \theta; \varphi)$$

We will **want to fit these variational parameters so that $q(\dots)$ approximates $p(\dots)$** .

To ease notation, let's lump together the cluster assignments and parameters:

$$z \equiv \{c, \pi, \mu\}$$

In the past, we used Z to denote the cluster assignments and referred to them as **latent variables** as the latent variables, while the cluster means and mixture weights were the **parameters**. As a rule-of-thumb the difference between latent variables and parameters is what you want to do with them: for latent variables, we want a posterior distribution, whereas for parameters we want a point estimate.

In this case, we want a posterior distribution over all variables, so **we will now refer to all of them as latent variables**. (Note that this is a norm which is inconsistently applied, both in this class, and in the broader literature.)

To recap, using simpler notation, we want the following:

$$q(z) \approx p(z | y)$$

Consider the optimization problem:

$$q^*(z) = \underset{q \in Q}{\operatorname{argmin}} KL(q(z) \parallel p(z | y))$$

For some family of densities Q , this will find the member that is closest in KL divergence to the exact posterior $p(Z | Y)$. This is the objective function for (the most commonly used form) of variational inference often known as **variational Bayes**.

How do you minimize the KL divergence to the intractable posterior? Let's perform a manipulation of the KL divergence (which should look familiar):

$$\begin{aligned} KL(q \parallel p) &= \mathbb{E}_q \left[\log \frac{q(z)}{p(z|x)} \right] \\ &= \underbrace{\mathbb{E}_q \left[\log \frac{q(z)}{p(z,x)} \right]}_{= -ELBO(q)} + \underbrace{\mathbb{E}_q \left[\log p(x) \right]}_{= \log p(x)} \\ &\quad \text{evidence lower bound (ELBO)} \quad \text{log evidence} \end{aligned}$$

$$\propto_q -ELBO(q)$$

So we can minimize the KL divergence to an intractable density by maximizing the ELBO:

$$\min_q KL(q \parallel p) = \max_q ELBO(q)$$

We saw this same fact when deriving EM. One difference here is that the **ELBO** is now just a functional of the q -distribution (rather than a functional of q and the parameters).

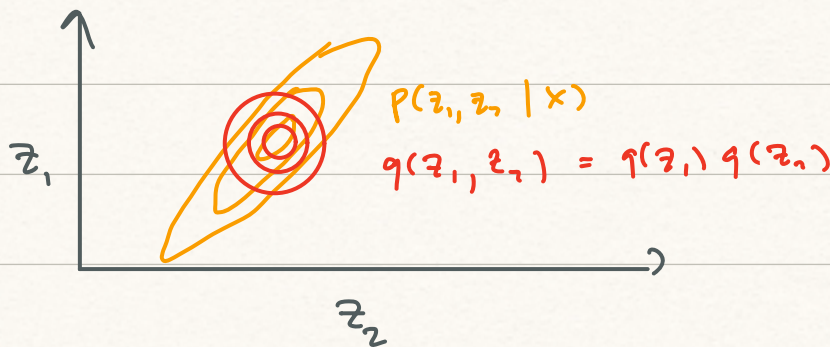
We now have a tractable objective function, but not necessarily a tractable way to optimize it. The next move that variational Bayes makes is to assume that $q(\dots)$ comes from a simple parametric family of densities that is easy to search over. Specifically, assume that $q(\dots)$ is a factorized family:

$$q(z_{1:D}) = \prod_d q(z_d)$$

In other words, all latent variables are marginally independent under the q -distribution. For example in the Bayesian mixture model:

$$\text{e.g. } q(c, \mu, \pi) = q(\pi) \left[\prod_j q(\mu_j) \right] \left[\prod_i q(c_i) \right]$$

This is called the **mean-field approximation**. Consider the simple picture below:



How does this assumption help us? It facilitates the following coordinate ascent algorithm

Coordinate ascent variational inference (CAVI)

until convergence:

for $d = 1 \dots D$:

$$q^*(z_d) \leftarrow \arg\max_{q(z_d)} \text{ELBO}(q)$$

Here is a fact from Bishop (2006): the maximizer of the ELBO takes the following form:

$$q^*(z_d) \propto \exp \left(\mathbb{E}_{q_{\cdot|z_d}} [\log p(z_d | z_{-d}, x)] \right)$$

Let's confirm it. Consider the KL divergence from any other setting of the density $q(z_d)$ to the optimal setting:

$$\text{KL}(q(z_d) \parallel q^*(z_d))$$

$$= \mathbb{E}_{q(z_d)} [\log q(z_d) - \log q^*(z_d)]$$

$$\propto \mathbb{E}_{q(z_d)} [\log q(z_d)] - \mathbb{E}_{q(z_d)} \left[\mathbb{E}_{q(z_{-d})} [\log p(z_d | z_{-d}, x)] \right]$$

^ by definition

$$\propto -H(q(z_d)) - \mathbb{E}_q [\log p(z, x)]$$

$$\propto -\text{ELBO}(q)$$

Therefore maximizing the ELBO wrt $q(z_d)$ would be minimizing the KL to $q^*(z_d)$, which is only achieved if $q(z_d) = q^*(z_d)$.

As we saw with EM, if the complete conditional $p(z_d | -)$ is exponential family, then the optimal setting $q^*(z_d)$ will be

$$p(z_d | -) = f(z_d; \eta(\dots)) \propto h(z_d) \exp(t(z_d)^T \eta(\dots))$$

where its posterior natural parameter will be a function of other latent variables and data.

$$q^*(z_d) = f(z_d; \mathbb{E}_{q_{\cdot|z_d}} [\eta(\dots)])$$

We will see this in practice next time.